

Gautham Krishna Gudur

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RESEARCH INTERESTS

I am interested in building data-efficient, data-centric, and human-centered machine learning systems.

Focus Areas: Foundation models, LLMs/SLMs, pre-training and post-training, {continual, active, few-shot} learning, token- and parameter-efficient learning (PEFT), reasoning, agentic AI, modality gaps, cross-modal alignment, personalization, adaptation.

Applications: Multimodal and multisensory learning (wearable, physiological, time-series, acoustic data), health AI, ubicomp.

EDUCATION

The University of Texas at Austin, Austin, TX

Aug 2023 – Present

Ph.D., Electrical and Computer Engineering, **GPA:** 3.9/4.0

Advisors: [Prof. Joydeep Ghosh](#) (prev. [Prof. Edison Thomaz](#))

Anna University, Chennai, India

2013 – 2017

B.Tech., Information Technology

SELECTED PUBLICATIONS

* denotes equal contribution and joint lead authorship

[Google Scholar Citations: 401, h-index: 10, i-index: 10]

Conference/Journal/Workshop

- RR Azghan, [Gautham Krishna Gudur](#), P Turaga, G Pedrielli, H Ghasemzadeh. [Latent-LoRA: Compact Latent-Space Adapters with Gradient-Free Routing for Continual Learning](#), **EMNLP 2026 Findings**.
- JH Park, [Gautham Krishna Gudur](#), Y Shen, D Liang, E Thomaz. [RAST: Resolution-Aware Privileged Structure Transfer for Low-Resolution Audio Activity Recognition](#), **ACM MobiCom 2026 – HumanSys** workshop.
- [Gautham Krishna Gudur*](#), M Malu*, T Khandait, RR Azghan, A Rayas, P Turaga, J Ghosh, et al. [PCL: Partitioned Continual Learning via Unsupervised Latent Experts for Audio Classification](#), **ICML 2026 – Machine Learning for Audio** workshop.
- R Azghan, [Gautham Krishna Gudur](#), M Malu, E Thomaz, G Pedrielli, P Turaga, H Ghasemzadeh. [Gated Adaptation for Continual Learning in Human Activity Recognition](#), **IEEE Internet of Things Journal 2026**.
- E Farahmand, RR Azghan, NT Chatrudi, E Kim, [Gautham Krishna Gudur](#), E Thomaz, G Pedrielli, P Turaga, H Ghasemzadeh. [AttenGlucO: Multimodal Transformer-Based Blood Glucose Forecasting on AI-READI Dataset](#), **IEEE EMBC 2025**.
- V Lingam*, A Tejaswi*, A Vavre*, A Shetty*, [Gautham Krishna Gudur*](#), J Ghosh, A Dimakis, E Choi, A Bojchevski, S Sanghavi. [SVFT: Parameter-Efficient Fine-Tuning with Singular Vectors](#), **NeurIPS 2024**.
Abridged versions: NeurIPS 2024 – FITML workshop; ICML 2024 – WANT [Oral Presentation] and ES-FoMo workshops.
- [Gautham Krishna Gudur](#), E Thomaz. [Dataset Distillation for Audio Classification: A Data-Efficient Alternative to Active Learning](#), **NeurIPS 2024 – ENLSP** workshop.
- G Tata*, [Gautham Krishna Gudur*](#), G Chennupati, ME Khan. [Can Calibration Improve Sample Prioritization?](#), **NeurIPS 2022 – HILL and HITY** workshops.
- [Gautham Krishna Gudur](#), R Raaghul, K Adithya, S Vasudevan. [Data-Efficient Automatic Model Selection in Unsupervised Anomaly Detection](#), **IEEE ICMLA 2022**.
- [Gautham Krishna Gudur](#), SK Perepu. [Zero-Shot Federated Learning with New Classes for Audio](#), **INTERSPEECH 2021**.
Abridged versions: ICLR 2021 – DPML and HAET workshops.
- [Gautham Krishna Gudur](#), SK Perepu. [Resource-Constrained Federated Learning with Heterogeneous Labels and Models for Human Activity Recognition](#), **IJCAI-PRICAI 2020 – DL-HAR, KDD 2020 – AIoT**, and **NeurIPS 2020 – MLMH** workshops.
- A Ragav*, [Gautham Krishna Gudur*](#). [Bayesian Active Learning for Wearable Stress and Affect Detection](#), **NeurIPS 2020 – MLMH** workshop.
- [Gautham Krishna Gudur](#), A Ramesh, R Srinivasan. [A Vision-based Deep On-Device Intelligent Bus Stop Recognition System](#), **ACM UbiComp 2019 – PURBA** workshop.
- [Gautham Krishna Gudur](#), P Sundaramoorthy, V Umaashankar. [ActiveHARNet: Towards On-Device Deep Bayesian Active Learning for Human Activity Recognition](#), **ACM MobiSys 2019 – EMDL** workshop.
- P Sundaramoorthy, [Gautham Krishna Gudur](#), MR Moorthy, RN Bhandari, V Vijayaraghavan. [HARNet: Towards On-Device Incremental Learning using Deep Ensembles on Constrained Devices](#), **ACM MobiSys 2018 – EMDL** workshop.

Preprints

- H Yoon, S Im, P Langer, T Lee, [Gautham Krishna Gudur](#), J Kim, X Han, YY Wu, N Zumarraga, Y Zhang, R Müller, J Ghosh, Y Yang, SJ Lee. [SensorHarness: Optimizing How LLMs Sense the World](#).

- X Han, Y Wang, C Chen, W Dei, [Gautham Krishna Gudur](#), S Li, HH Chung, GD Hager, J Ghosh, P Liang, S Saria. Counterfactual Self-Evolution for Evidence-Grounded Reasoning.
- RR Azghan, [Gautham Krishna Gudur](#), M Malu, E Thomaz, G Pedrielli, P Turaga, H Ghasemzadeh. [CLAD-Net: Continual Activity Recognition in Multi-Sensor Wearable Systems](#).
- E Farahmand, RR Azghan, NT Chatrudi, VYA Baidoo, E Kim, [Gautham Krishna Gudur](#) et al. [GluMind: Multimodal Parallel Attention and Knowledge Retention for Robust Cross-Population Blood Glucose Forecasting Systems](#).

RESEARCH AND WORK EXPERIENCE

The University of Texas at Austin – Graduate Research Assistant

Austin, TX

Advisors: [Prof. Joydeep Ghosh](#), [Prof. Edison Thomaz](#)

Aug 2023 – Present

- Devised test-time LLM user personalization for multimodal time-series health tasks using few-shot in-context learning.
- Proposed SVFT (Singular Vectors guided Fine-Tuning), a Pareto-optimal PEFT technique [[Mentor: Prof. Sujay Sanghavi](#)].
- Designed continual learning methods for audio sensing, multi-sensor HAR, and cross-population glucose prediction using unsupervised latent partitioning and LoRA adapters; reduced labeling costs up to 3000× via dataset distillation.
- Developed multimodal and cross-modal alignment techniques for time-series, language, health, and audio.

Bosch Research – Foundation Models Research Intern (BCAI)

Sunnyvale, CA

Mentors: [Gorden Li](#), [Wenbin He](#), [Liu Ren](#)

May 2026 – Aug 2026

- Worked at the intersection of foundation models, SLM/LLM adaptation, and multimodal task-agnostic time-series Q&A.

Nokia Bell Labs – Research Intern (Device Intelligence team)

Cambridge, UK

Mentors: [Soumyajit Chatterjee](#), [Mohammad Malekzadeh](#), [Fahim Kawsar](#)

May 2025 – Aug 2025

- Improved generalizability of PPG-based health foundation models using test-time adaptation.

Independent Research

Dec 2018 – May 2023

- Analyzed the effect of calibration on prioritizing important samples during training [[Mentor: Prof. Emtiyaz Khan](#), [RIKEN](#)].
- Worked on Bayesian active learning for on-device human activity recognition (HAR) and wearable sensing tasks.

Ericsson R&D, Global AI Accelerator (GAIA) – Data Scientist III

Chennai, India

Mentors: [Shrihari Vasudevan](#), [M J Prasath](#)

Feb 2019 – Apr 2023

- Worked on ML for network intelligence and telecom, leading to multiple publications, patents, and AI-driven solutions.
- Advanced [Ericsson's AI-Native](#) telecom stack: 3GPP standardization for federated learning and multi-vendor model sharing, 5G NWDAF mobility and connectivity modeling, and unsupervised Bayesian anomaly detection ([E-ADF](#)).
- Built telco-AI platforms: Ericsson LLM Hub with domain-rich data and LLMs, language translation, active learning for customer-symptom modeling, and object detection for automated detection of infrastructure failures at cell sites.
- Designed zero-shot federated learning frameworks to handle new heterogeneous classes and models.

SmartCardia (EPFL) – Machine Learning Engineer

Chennai, India

Mentor: [Srinivasan Murali](#)

May 2018 – Nov 2018

- Implemented gradient-boosted ensembles and LSTM models for regression and classification on imbalanced time-series wearable clinical data; engineered biomarker-driven features for sleep apnea, troponin, blood pressure, and hemoglobin.

Solarillion Foundation – Research Assistant

Chennai, India

Mentor: [Vineeth Vijayaraghavan](#)

Feb 2016 – May 2018

- Led the development of [HARNet](#) – deep ensemble models for activity recognition with on-device incremental learning.
- Designed dynamic gesture recognition models with efficient feature engineering on a low-cost Raspberry Pi Zero (\$5).

SSN College of Engineering – Undergraduate Student Researcher

Chennai, India

Mentors: [Prof. Bhalaji N](#), [Prof. Srinivasan R](#)

Feb 2015 – Mar 2017

- Developed a CNN-based bus stop recognition system with ~96% accuracy in dynamic Indian environments.
- Led an HCI research project on Neurocinematics to classify real-time cognitive responses of film viewers using EEG data.

PATENTS

- Efficient Backpropagation Free Test-Time Adaptation on Foundation Models for PPG [[Filed](#)].
- [Federated Learning using Heterogeneous Labels](#), WO2022013879A1.
- [Distributed Machine Learning with New Labels using Heterogeneous Label Distribution](#), WO2022162677A1.
- [Method and Apparatus for Approach Recommendation with Threshold Optimization in Unsupervised Anomaly Detection](#), WO2023166515A1.

SELECTED TALKS

- Resource-efficient Machine Learning – SSN College of Engineering (2022).
- [Heterogeneous Zero-Shot Federated Learning with New Classes for On-Device Audio Classification](#) – MobiUK 2021.

- Telecom-Specific Language Translation using GCP – Ericsson/Google Cloud Day (2021).
- [Resource-Constrained Machine Learning for Ubiquitous Computing Applications](#) – Flipped by GAIUS (2020).

SELECTED HONORS, AWARDS, AND FELLOWSHIPS

- [SVFT](#) was selected for an **oral presentation** at the WANT workshop at ICML 2024.
- *Graduate Ph.D. Fellowship* from the Cockrell School of Engineering at The University of Texas at Austin.
- [Top 1 percentile on HackerRank](#) in the Algorithms Domain – Problem Solving (Advanced).
- Full financial registration grants to attend ICLR 2021, NeurIPS 2020, and OxML 2020.
- *AIB* won Ericsson's Top Performance Competition 2020 in the Operational Excellence category.
- Undergraduate research grant of INR 25,000 from SSN College of Engineering.
- Winner of the GermEval Shared Task 1 Challenge (Subtask (a)), KONVENS 2019 in the post-evaluation phase.

PROFESSIONAL SERVICE

- **Organizing Committee:** [WellComp 2026](#) workshop at ACM UbiComp/ISWC 2026.
- **Program Committee/Reviewer:** IEEE ICASSP 2026/2025, Interspeech 2026, IEEE MLSP 2026, ACM UbiComp/ISWC 2025, ACM ICMI 2025, ICML 2024 – ES-FoMo workshop, ICLR 2021 – DPML workshop, NeurIPS – ML4H 2020/2019.
- Event Organizer: Data Nuggets – a data science contest at Invente 2016, SSN College of Engineering.

TEACHING AND MENTORING

- **Graduate Teaching Assistant, UT Austin** – *Primary Instructors:* [Prof. Joydeep Ghosh](#), [Prof. Edison Thomaz](#)
[Applied Machine Learning](#) (Fall 2026), [Deep Learning](#) (Fall 2025), and [Human Signals](#) (Spring 2025) graduate courses.
Designed assignments, exams, and quizzes; mentored students on paper reviews, presentations, and projects.
- Aadharsh Aadithya (UT Austin) – Discrete diffusion models for time-series data (2026 – Ongoing).
- Ji Hwan Park (UT Austin) – Audio sensing and multimodal alignment (2025 – 2026).
- Abhijith Ragav (GaTech → Amazon) – Bayesian active learning for wearable stress and affect detection (2020).
- Prahalathan Sundaramoorthy (USC → HP Labs) – Bayesian active learning for human activity recognition (2018).
- Manav Rajiv Moorthy (UWash → Amazon) – Deep learning and human activity recognition (2017 – 2018).
- TA, Solarillion Foundation – Mentored 10+ students on embedded ML research projects/assignments (Feb '17 - May '18).

SUMMER SCHOOLS

- **5th Summer School on Artificial Intelligence** 2021, IIIT Hyderabad.
- **Eastern European Machine Learning Summer School (EEML)** – both 2020 and 2021
Presented ActiveHARNet (EEML '20); zero-shot federated learning and task-independent continual learning (EEML '21).
- **Oxford Machine Learning Summer School (OxML)** 2020.

POSTERS AND EXTENDED ABSTRACTS

- Uncertainty-Aware Chain-of-Thought Reasoning in Large Language Models, AI in Health course.
- [Adaptive Federated Learning in Conceptually Drifting Environments](#), Applied Machine Learning course.
- [Heterogeneous Zero-Shot Federated Learning with New Classes for On-Device Audio Classification](#), **MobiUK 2021**.
- [Bayesian Active Learning for Wearable and Mobile Health](#), **NeurIPS** Europe meetup on Bayesian Deep Learning (**BDL 2020**).
- [Handling Real-time Unlabeled Data in Activity Recognition using Deep Bayesian Active Learning and Data Programming](#), **MobiUK 2019**, University of Oxford.
- [Neurocinematics: The Intelligent Review System](#), 3rd International Conference on Cognition, Brain and Computation (**CBC 2015**), IIT Gandhinagar.

TECHNICAL COURSEWORK AND MOOCs

- **Relevant Ph.D. Coursework**
Generative Models in Machine Learning, Advanced Computer Vision, Statistical Methods I & II, Ethics of AI, Applied Machine Learning, Fine-Tuning LLMs, AI in Health, Spoken Language Technologies, Human Signals
- HackerRank | [Problem Solving](#) – Advanced, Intermediate, Basic
- University of Washington | Coursera – [Machine Learning Specialization](#) (4 courses)
A Case Study Approach, Regression, Classification, Clustering & Retrieval

TECHNICAL SKILLS

Programming Languages & Tools: Python, C++, HTML/CSS, Bash, SQL, JavaScript, Git, LaTeX, Claude Code, Docker, GCP

Libraries & Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, Keras, Numpy, Pandas, Scikit-learn, PySpark